DEPARTMENT OF THE AIR FORCE
HEADQUARTERS 341ST SPACE WING (AFSPC)

14 June 2001

341 CS/SCBR (FOIA)
30 73d Street North
Malmstrom AFB MT 59402-7513

Mr. John Greenewald, Jr.
[redacted]

Dear Greenewald

On 18 May 2001, we received your Freedom of Information Act (FOIA) request dated 9 May 2001, for records pertaining to a March 16, 1967 incident, in which there was a malfunction of missile silos. Copies of excerpts, as well as supporting documents, of the below listed documents are releasable and are attached:

a. 341st Strategic Missile Wing History (1 Jan – 31 Mar 1967)
b. 341st Strategic Missile Wing History (1 Apr – 30 Jun 1967)
c. 341st Strategic Missile Wing History (1 Jul – 30 Sep 1967)
d. 341st Strategic Missile Wing History (1 Oct – 31 Dec 1967)
e. 341st Strategic Missile Wing History (1 Jan – 31 Mar 1968)

We do not have any audio or video tapes, or documents of any personnel injured. If you are pursuing your inquiry further, you may wish to contact the National Archives and Records Administration (NARA) at:

NARA FOIA Officer
National Archives at College Park
8601 Adelphi Road, Room 4400
College Park MD 20740-6001
Web Site: www.nara.gov/foia

THIS EXCERPT IS:

[redacted]

K-WG-341-HI

341st STRATEGIC MISSILE WING

AND 341st COMBAT SUPPORT GROUP

HQ SAC DXIH 67-1865

(Unclassified Title)

1 Jan thru 31 Mar 1967

PRC

GROUP-3

Assigned to the

FIFTEENTH AIR FORCE, STRATEGIC AIR COMMAND

Permanently Stationed at

MALMSTROM AIR FORCE BASE, GREAT FALLS, MONTANA

PRC

DOCUMENT DECLASSIFIED PER HQ AFSPC/DOXN 27 Apr 95 and OO-ALC/LME 27 Jul 95.

This document was prepared by A2C David B. Gamble, Wing Historian under the supervision of the Wing Information Officer. It is prepared in compliance with SACR 210-1, 29 June 1964, and is classified SECRET under the provision of AFR 205-1 as amended. The classification is required for the protection of information that reveals the military capability and operational status of the Wing. (U)

APPROVED:

HERMAN T. DeHAAS, Capt, USAF
Chief, Information Division

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Wing Commander

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Project How Now (Service Star), the testing of Mark 5 re-entry vehicles by higher headquarters for effectiveness. Malmstrom sent the last vehicle in the program for the base during the quarter. This ended the program at Malmstrom.[58] (U)

A walk-thru inspection of the HF Hardened Antennas at Malmstrom was conducted during February by Boeing and SATAF personnel. The antennas were already equipped in the LFs but were not in working order. The walk-thru inspection had an average of 40 descrepencies per site. Boeing Company has not set any offical date for an operational antenna to be demonstrated. The estimated date for the operational antennas was set for early July 67. At that time, maintenance will begin on correcting the discrepencies at the sites.[59] (U)

The following is the investigation of Echo Flight incident and the results. (U)

On 16 March 1967 at 0845, all sites in Echo (E) Flight, Malmstrom AFB, shutdown with No-Go indication of Channels 9 and 12 on Voice Reporting Signal Assemble (VRSA). All LF's in E Flight lost strategic alert nearly simultaneously. No other Wing I configuration lost strategic alert at that time.[60] ([redacted])

Guidance & Control channel 50 dump data was collected from E-7 facility and E-8 Facility and all 10 sites were then returned to strategic alert without any LF equipment replacement. All 10

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58. Taken from Messages on file at Job Control, by A2C David B. Gamble, Wing Historian, on 23 Apr 67.

59. Rpt, "Monthly Report HF/SSB Hardened Antenna," by SATAF (DCA), 28 Feb 67, Ex 42. [redacted]

60. Msg, (S) DM 02752, SAC to 341 SMW, Subj: Loss of Strategic Alert Echo Flight, Malmstrom AFB, 17 Mar 67, Ex 43.

[redacted]

sites were reported to have been subject to a normal controlled shutdown.[61] [redacted]

The only unusual LF events noted were the failure of the secondary door actuator motor at LF, E-2 and the intermittent operation of the diesel generator at LF, E-8.[62] [redacted]

Technial Analysis Division (TAD) personnel inspected LCC, E-1 for loose cable connections and the Electric Surge Arrester (ESA) room for any faults and noted no unusual discrepancies.[63] (U)

LCF, E-1 experienced commercial power problems in the early afternoon of 16 Mar 67 which resulted in a burn out of the 10 Hp Envirmental Control System (ECS) chiller compressor motor.[64] (U)

At approximately 1400 hrs on 16 Mar 67, the 341st SMW was advised that SAC Headquarters had called in OOAMA support for a complete engineering analysis of this problem.[65] [redacted]

OOAMA decided to send a task group to Malstrom for study of the incident at Echo Flight because the problem pertained peculiarly to Wing I. It was also decided to make the studies and tests there because OOAMA was not equiped with Wing I equipment. The task group included personnel from OOAMA, Boeing Company, Autonetics, and 15th AF ; they were to arrive on 17 or 18

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61. Rpt, (S) "Report of Engineering Investigation of Echo Flight Incident, Malmstrom AFB, Mont. - 16 Mar 67," by Engineering Investigation Team, 23 Mar 67, p 1.

62. Ibid.

63. Ibid.

64. Ibid.

65. Ibid.

[redacted]

[redacted]

March 67.[66] (S)

A preliminary analysis was made of the Fault Isolator Test Tape (FITT) from LF, D-7 and E-8. Targeting support was schedualed for a G&C Channel 50 data dump at LF's E-2 and E-9 which, although returned to alert, were still felt to possess useful data. The investigation was held in suspense until the arrival of the OOAMA team on 18 March 67.[67] (S)

Echo Flight incident was approached in four ways in the investigation:[68] (S)

a. Review of events on or near 16 March 67 and of flight configuration.

b. Investigation, and where possible elimination, of circumstances which may have been responsible for the incident.

c. Investigation of means of causing the results which were noted at the time of shutdown.

d. Investigation of similar events.

In reviewing the events of the incident, the LCC crew was questioned by Wing Maintenance Evaluation Team (DOMET), OOAMA, and Boeing Company personnel on 16 March 67.[69] (S)

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66. Msg, (C) OONE 01012, OOAMA to SAC, Subj: Wing I, E Flight Incident, 3 Mar 67, Ex 44.

67. Rpt, (S) "Report of Engineering Investigation of Echo Flight Incident, Malmstrom AFB, Mont. - 16 Mar 67," by Engineering Investigation Team, 23 Mar 67, p 1.

68. Ibid., p 3.

69. Ibid., p 6.

[redacted]

[redacted]

In screening the crew it was determined that loss of strategic alert and fault indication occurred in an extremely short period of time. The exact time estimates could not be obtained but the time interval was estimated to be between 10 to 40 seconds.[70] [redacted]

The standby light was not observed to illuminate for any of the ten launch facilities. None of the command and status line monitor fault lights were illuminated. There was no pattern or sequence noted as to the illumination of fault and loss of strategic alert. No flickering or ripple was noted on any of the other status displays. The emergency lights were on (all other lights off) and these lights did not appear to dim.[71] [redacted]

The initial time of the incident, decided by the crew, could be no more than two or three minutes earlier than the official log time of 0845 hours (local).[72] [redacted]

The crew also stated; the LF, E-8 had a channel 26 which indicated it was operating on diesel at the time of fault. The 465L (SAC Communications & Control System) and communications equipment was not processing traffic at the time of the incident.[73] [redacted]

Neither of the crew members were using any electrical equipment. Such as electric razor...etc. At the time of fault

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70. Rpt, (S) "Report of Engineering Investigation of Echo Flight Incident, Malmstrom AFB, Mont. - 16 Mar 67," by Engineering Investigation Team, 23 Mar 67, p 6.

71. Ibid.

72. Ibid. [redacted]

73. Ibid., p 7.

the deputy commander was turned away from the console to brief the commander who had just been awakened from sleep. The commander was the first to see the faults.[74] (S)

Channel 50 data was extracted from sites E-7 and E-8 immediately after the shutdown of the entire flight. Analysis of this data determined that both sites were shutdown as a result of external influence to the G&C, no No-Go's were deltected by the G&C. Althouth E-8 had indications of a previous Monitor check No-Go it was not the cause of shutdown at this time. This was determined by information contained in sector 54 which said that subsequent to the last No-Go the G&C had completed an iteration routine which is performed approximately 2 hours and 30 minutes into a restart. Had a G&C No-Go caused this shutdown, the Discrete Input "A" (DIA) word would have been placed in this sector.[75] (U)

On Sunday, 19 March 67, a team was dispatched to sites E-2 and E-9 to dump channel 50 data. These two sites were in strategic alert at this time but were selected because they had never experienced a restart since the last time that a maintenance tape had been used. Information from the No-Go sectors of channel 50 indicated that these G&C's had never entered a No-Go loop at any time. This hardens the case against a No-Go causing a shutdown of the flight. If a No-Go had shutdown the flight, it would have

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74. Rpt, (S) "Report of Engineering Investigation of Echo Flight Incident, Malmstrom AFB, Mont. - 16 Mar 67," by Engineering Investigation Team, 23 Mar 67, p 7.

75. Ibid.,p 8.

[redacted]

been recorded in channel 50 data.[76] (U)

The only non-detectable fault from the G&C section which could have caused a No-Go would have been loss of confirm codes. This type of fault would not have produced a VRSA channel 9 and 12, but only a channel 9. This information eliminated the No-Go theory as the cause of the incident.[77] (U)

As stated eariler, all 10 launch facilities shutdown with a VRSA channel 9 and 12 (G&C No-Go and Logic Coupler No-Go) recordings. Because of this consistency considerable investigation was expended in the Logic Coupler area. In the channel 50 analysis it was shown that the guidance section did not experience a No-Go and therefore, it was felt that the VRSA channel 9 report was not valid. It is possible, however, for the Logic Coupler to generate both of these No-Go indications.[78] [redacted] (U)

The logic of the coupler was studied by the investigating team in an effort to identify a method by which both VRSA 9 and 12 could be activated. The opinion of the team was that external generated signals caused the generation of these two channels and shutdown of the launch facilities. The possibility of this is very remote due to the fact that all 10 couplers would have to fail in the flight within a few seconds of each other.[79] [redacted] (U)

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76. Rpt, (S) "Report of Engineering Investigation of Echo Flight Incident, Malmstrom AFB, Mont. - 16 Mar 67, by Engineering Investigation Team, 23 Mar 67, p 8.

77. Ibid., p 30.

78. Ibid., p 30. [redacted]

79. Ibid., p 30.

[redacted]

The only possible means that could be identified by the team involved a situation in which a coupler self test command occurred along with a partial reset within the coupler. This could feasible cause a VRSA 9 and 12 indication. This was also quite remote for all 10 couplers would have to have been partially reset in the same manner.[80] [redacted] (U)

Further studies of this problem will be accomplished at the contractors facility since a full engineering investigation is not feasible at this level. (U)

In the researching of other possibilities, weather was ruled out as a contributing factor in the incident.[81] (U)

A check with Communication maintenance verified that there was no unusual activity with EWO-1 or EWO-2 at the time of the incident.[82] (U)

Rumors of Unidentified Flying Objects (UFO) around the area of Echo Flight during the time of fault were disproven. A Mobile Strike Team, which had checked all November Flight's LFs on the morning of 16 March 67, were questioned and stated that no unusual activity or sightings were observed.[83] (U)

The 801st Radar Squadron, Malmstrom AFB, gave a negative report on any radar or atmospheric interference problems related to Echo Flight.[84] (U)

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80. Rpt, (S) "Report of Engineering Investigation of Echo Flight Incident, Malmstrom AFB, Mont - 16 Mar 67," by Engineering Investigation Team, 23 Mar 67, p 30.

81. Ibid., p 28.

[redacted]

82. Ibid., p 25.
83. Ibid., p 27.

84. Ibid., p 26.

[redacted]

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Capt. Bradshaw in the command post at the time of Echo Flight incident, verified that there was no activity on the Primary Alerting System (PAS) at Echo Flight.[85] (U)

During the investigation it was discovered that Boeing modification teams were at sites B-1, K-1 and G-1 on 16 March 67; but this could not have affected anything in relationship to E Flight because they arrived at the sites after the incident occurred.[86] (U)

As mentioned before, a power problem affected E-1 at 1453 on 16 March 67 according to LCC log. The Brine Chiller (CH-1) shutdown due to power fluctuation. This was not considered unusual for power failures or fluctuations often cause CH shutdowns. Between 1510 and 1600 hours, E-1 experienced another power fluctuation. The capsule crew heard the Motor Generator brushes chatter. At this time the site manager indicated that one phase had low voltage, but LCF power was not appreciably affected since capsule florescent lights were still on. The diesel generator started but could not be put on the line due to broken lead. Commercial power was restored at 1921 hours on 16 March 67.[87] (U)

In reviewing the maintenance history of the Wing, it was discovered a similar incident occurred at Alpha Flight in

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85. Rpt, (S) "Report of Engineering Investigation of Echo Flight Incident, Malmstrom AFB, Mont - 16 Mar 67," by Engineering Investigation Team, 23 Mar 67, p 26.

86. Ibid., p 23.

87. Ibid., p 12.

[redacted]